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REINFORCING BRACKET FOR BATTERY, BATTERY, AND POWER CONSUMING APPARATUS

Abstract

A battery includes a plurality of battery cells, the reinforcing bracket includes a connecting bracket and a plurality of partition members, the connecting bracket includes a first reinforcing plate, a plurality of avoidance holes that are spaced apart along a first direction are provided on the first reinforcing plate, the plurality of partition members are spaced apart along the first direction on the connecting bracket, and a placement space is defined between the adjacent partition members and the connecting bracket.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS [0001] The present application is a continuation of International Application No. PCT/CN2023/132384, filed on Nov. 17, 2023, which claims priority to Chinese Patent Application No. 202321086493.6, filed with the China National Intellectual Property Administration on May 8, 2023 and entitled “REINFORCING BRACKET FOR BATTERY, BATTERY, AND POWER CONSUMING APPARATUS”, each are incorporated herein by reference in their entirety.

TECHNICAL FIELD

[0002] The present application belongs to the technical field of batteries, and specifically, to a reinforcing bracket for a battery, a battery, and a power consuming apparatus.

BACKGROUND

[0003] A battery generally includes a box and a plurality of battery cells disposed in the box.

[0004] In the related art, to ensure structural strength between adjacent battery cells, a plurality of limiting members configured to limit the battery cells are generally disposed in the box.

[0005] However, currently, structural strength of the limiting members is relatively low, and a large space is occupied due to cooperation between the plurality of limiting members. Consequently, the limiting members cannot be configured to effectively limit the battery cells, and an energy density of the battery is further reduced. Therefore, a limiting member that has high structural strength and occupies a small space is urgently needed.

SUMMARY

[0006] To this end, a first aspect of this application provides a reinforcing bracket for a battery, to effectively limit a plurality of battery cells and ensure an energy density of the battery.

[0007] According to the reinforcing bracket for a battery in embodiments of this application, the battery includes a plurality of battery cells, and the reinforcing bracket includes: a connecting bracket, where the connecting bracket includes a first reinforcing plate, a plurality of avoidance holes that are spaced apart along a first direction are provided on the first reinforcing plate, each of the avoidance holes runs through the first reinforcing plate in a second direction, and the first direction is perpendicular to the second direction; and a plurality of partition members, where the plurality of partition members are spaced apart on the connecting bracket along the first direction, and a placement space for placing the battery cells is defined between the adjacent partition members and the connecting bracket.

[0008] According to the reinforcing bracket for a battery in embodiments of this application, the first reinforcing plate is disposed on the reinforcing bracket, to improve structural strength of the reinforcing bracket, so as to ensure that the battery cells can be stably limited by the reinforcing bracket, thereby improving position stability of the battery cells and ensuring structural strength between the adjacent battery cells. In addition, the reinforcing bracket is disposed to include the plurality of partition members that are spaced apart along the first direction. The placement space is formed on the reinforcing bracket through cooperation between the plurality of partition members and the connecting bracket, and the battery cells are placed in the placement space. At least one

battery cell can be limited by the reinforcing bracket, to resolve the technical problem that a space for disposing the battery cells is reduced due to disposition of the plurality of reinforcing brackets, thereby improving the energy density of the battery. In addition, the battery cells can be prevented from being displaced in the first direction through cooperation between the plurality of partition members, thereby stably limiting the battery cell by the reinforcing bracket, and further improving the position stability of the battery cells.

[0009] In some embodiments, the reinforcing bracket further includes second reinforcing plates, where the second reinforcing plates are disposed on at least one side of each placement space in a third direction, and the third direction is separately disposed perpendicular to the first direction and the second direction. The second reinforcing plates not only can improve the structural strength of the reinforcing bracket, but also can directly limit the battery cells, thereby improving the position stability of the battery cells in the placement space.

[0010] In some embodiments, the second reinforcing plates are disposed on two sides of each placement space in the third direction, to further improve the structural strength of the reinforcing bracket. In addition, the battery cells may be further limited through cooperation between the two second reinforcing plates, to avoid displacement of the corresponding battery cell in the third direction, thereby ensuring the structural strength between the adjacent battery cells.

[0011] In some embodiments, the second reinforcing plates separately protrude from two sides of the same partition member in the first direction. In this way, one second reinforcing plate may be simultaneously disposed on side portions of two adjacent placement spaces, so that two battery cells are simultaneously limited by one second reinforcing plate. In this way, production costs and manufacturing difficulty of the reinforcing bracket can be reduced while the position stability of the battery cell is improved.

[0012] In some embodiments, a height of the second reinforcing plate is less than a height of the partition member in the second direction. While ensuring the reinforcing effect of the second reinforcing plates, materials of the second reinforcing plates may be further reduced, so as to reduce manufacturing costs of the reinforcing bracket and further reduce difficulty in forming the reinforcing bracket.

[0013] In some embodiments, the second reinforcing plates and the connecting bracket are spaced apart, to further reduce difficulty in forming the reinforcing bracket, and processing efficiency of the reinforcing bracket is improved.

[0014] In some embodiments, a through-hole is provided on each partition member, and the through-hole runs through the partition member in the first direction. While the manufacturing costs of the partition member are reduced by setting the through-hole, it is further conducive to imply a fixed connection between the adjacent battery cells, thereby improving connection strength of the adjacent battery cells. In addition, heat dissipation of the battery cells may be further accelerated, and normal expansion between the adjacent battery cells may be ensured, thereby improving service safety of the battery cells.

[0015] In some embodiments, the connecting bracket includes an annular enclosing plate, the enclosing plate is connected to an outer edge of the first reinforcing plate, and the enclosing plate extends along the second direction. In addition, while the structural strength of the reinforcing bracket is improved, the battery cells may be further limited by the enclosing plate, to avoid displacement of the battery cells, thereby ensuring the structural strength between the adjacent battery cells.

[0016] In some embodiments, the reinforcing bracket is symmetrically disposed along a first center line extending parallel to the first direction. In this way, the reinforcing bracket is formed as a symmetric member that is symmetrically disposed along the first center line, to reduce processing difficulty of the reinforcing bracket, and to ensure uniform limiting strength of the reinforcing bracket on the plurality of battery cells, and improve the position stability of the plurality of battery cells.

[0017] In some embodiments, the reinforcing bracket is symmetrically disposed along a second center line parallel to the second direction. In this way, the reinforcing bracket is formed as a symmetric member that is symmetrically disposed along the second center line, to further reduce processing difficulty of the reinforcing bracket, and to ensure uniform limiting strength of the reinforcing bracket on the plurality of battery cells, and improve reliability of the entire battery.

[0018] According to a second aspect, this application provides a battery, including a reinforcing bracket, where the reinforcing bracket is the reinforcing bracket in the foregoing embodiments; and a plurality of battery cells, where the battery cells are placed in each placement space, the first reinforcing plate is located on an outer side of the corresponding battery cell, and an electrode terminal of each battery cell passes through a corresponding avoidance hole.

[0019] In the technical solutions of the embodiments of this application, the reinforcing bracket in the foregoing embodiments is used, so that structural stability and reliability of the battery can be effectively ensured, and an energy density of the battery can be improved.

[0020] In some embodiments, an adhesive layer is disposed between the reinforcing bracket and each of the battery cells. The battery cells are fixed in the placement space of the reinforcing bracket by the adhesive layer, to improve connection strength between the battery cells and the reinforcing bracket, thereby making relative positions between the reinforcing bracket and the battery cells stable, and achieving an objective of limiting the battery cells and reinforcing a structure by the reinforcing bracket.

[0021] In some embodiments, the battery further includes a box, where the plurality of battery cells are disposed in the box, and the reinforcing bracket is fixed to the box. The reinforcing bracket and the plurality of battery cells are supported by the box, to improve relative position stability between the reinforcing bracket and the plurality of battery cells.

[0022] According to a third aspect, this application provides a power consuming apparatus, including a battery, where the battery is the battery in the foregoing embodiments, and the battery is configured to provide electric energy.

[0023] In the technical solutions of the embodiments of this application, the battery in the foregoing embodiments is used, that is, the reinforcing bracket in the foregoing embodiments is used, so that structural stability of the power consuming apparatus can be effectively improved and structural strength of the power consuming apparatus can be ensured.

[0024] Additional aspects and advantages of the present application will be given in the following descriptions, some of which will become apparent from the following descriptions or may be learned from practices of the present application.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0025] The foregoing and/or additional aspects and advantages of the present application will become apparent and comprehensible in the embodiment description made with reference to the following accompanying drawings.

[0026] FIG. 1 is a schematic diagram of a battery according to some embodiments of the present application;

[0027] FIG. 2 is a schematic diagram of a reinforcing bracket cooperating with a plurality of battery cells according to some embodiments of the present application;

[0028] FIG. 3 is a schematic diagram of a reinforcing bracket according to some embodiments of the present application;

[0029] FIG. 4 is a partial enlarged view of a region I in FIG. 3;

[0030] FIG. 5 is a top view of a reinforcing bracket according to some embodiments of the present application;

[0031] FIG. 6 is a front view of a reinforcing bracket according to some embodiments of the present application; and

[0032] FIG. 7 is a schematic diagram of a power consuming apparatus according to some embodiments of the present application.

REFERENCE NUMERALS

[0033] **1000**. Power consuming apparatus; [0034] **200**. battery; [0035] **10A**. battery unit; [0036] **10**. battery cell; [0037] **11**. side wall; **111**. first side wall; **112**. second side wall; [0038] **12**. electrode terminal; [0039] **30**. box; [0040] **120**. reinforcing bracket; [0041] **121**. connecting bracket; [0042] **1211**. first reinforcing plate; **121a**. avoidance hole; [0043] **1212**. enclosing plate; [0044] **122**. partition member; **1221**. through-hole; [0045] **123**. second reinforcing plate; [0046] **124**. placement space; [0047] X. first direction; Y. second direction; and Z. third direction.

DETAILED DESCRIPTION

[0048] To make the objectives, technical solutions, and advantages of embodiments of this application clearer, the following clearly describes the technical solutions in the embodiments of the present application with reference to the accompanying drawings in the embodiments of the present application. Apparently, the described embodiments are merely some rather than all of the embodiments of the present application. All other embodiments obtained by a person of ordinary skill in the art based on embodiments of the present application without making creative efforts shall fall within the protection scope of the present application.

[0049] Unless otherwise defined, meanings of all technical and scientific terms used in the present application are the same as those commonly understood by a person skilled in the art to which the present application belongs; and the terms used in the descriptions of the present application are merely for the purpose of describing specific embodiments, but are not intended to limit the present application. The terms “comprise” and “have” and any variations thereof in the specification and the claims of the present application as well as the descriptions of the accompanying drawings are intended to cover non-exclusive inclusion. The terms “first”, “second” and the like in the specification and claims of the present application or the above accompanying drawings are used to distinguish different objects, rather than to describe a specific order or primary-secondary relationship.

[0050] “Embodiment” mentioned in this application means that specific features, structures, or characteristics described with reference to the embodiment may be included in at least one embodiment of the present application. The term appearing at different positions of the specification may not refer to the same embodiment or an independent or alternative embodiment that is mutually exclusive with another embodiment. A person skilled in the art explicitly or implicitly understands that the embodiments described in the specification may be combined with other embodiments.

[0051] In the descriptions of the present application, it should be noted that, unless otherwise expressly specified and limited, the terms “mount”, “connect”, “connection” and “attachment” should be understood in a broad sense. For example, the connection may be a fixed connection, a detachable connection, or an integral connection; or the connection may be a direct connection, an indirect connection through an intermediate medium, or internal communication between two elements. A person of ordinary skill in the art may understand the specific meanings of the foregoing terms in the present application according to specific situations.

[0052] The term “and/or” in the present application only describes an association relationship for describing associated objects and represents that three relationships may exist. For example, A and/or B may represent the following three cases: Only A exists, both A and B exist, and only B exists. In addition, the character “/” in the present application generally indicates an “or” relationship between the associated objects.

[0053] In the embodiments of the present application, the same reference numerals denote the same components, and for the sake of brevity, the detailed description of the same components is omitted

in different embodiments. It should be understood that the dimensions, such as thickness, length and width, of the various components in the embodiments of the present application illustrated in the accompanying drawings, as well as the dimensions, such as an overall thickness, length and width, of an integrated apparatus are merely illustrative and should not be construed to limit the present application in any way.

[0054] In the present application, “a plurality of” means two or more (including two).

[0055] Nowadays, from the perspective of development of the market situation, batteries **200** are increasingly applied. Batteries **200** are not only used in energy storage power systems such as a water power plant, a fire power plant, a wind power plant, and a solar power plant, but also in electric transportations such as electric bicycles, electric motorcycles, electric vehicles, as well as in military equipment, aerospace and other fields.

[0056] With continuous expansion of application fields of the battery **200**, market demands for the battery **200** are also expanding.

[0057] In a manufacturing process of the battery **200**, a battery cell **10** is a basic element forming the battery **200**. A plurality of battery cells **10** are generally arranged in a box **30** in a specific order. In addition, during arrangement of the battery cells **10**, the battery cells **10** need to be limited by a limiting member, to limit positions of the battery cells **10**, so as to improve position stability of the battery cells **10** and ensure structural strength between adjacent battery cells **10**, thereby finally forming the battery **200** having relatively high structural strength and specific reliability.

[0058] However, the applicant notes that in the existing battery **200**, structural strength of the limiting member is relatively poor, and the battery cells **10** cannot be effectively limited. Consequently, a phenomenon that the battery cells **10** shake occurs. In addition, a plurality of limiting members need to be disposed for the plurality of battery cells **10** in the box **30**. As a result, a capacity for deploying the battery cells **10** in the box **30** is reduced, and an energy density of the battery **200** is reduced.

[0059] To resolve the foregoing problems, the embodiments of this application provide a reinforcing bracket **120** that can improve the energy density of the battery **200** and effectively limit the battery cells **10**. A specific solution is that a first reinforcing plate **1211** is disposed on a connecting bracket **121** of the reinforcing bracket **120**, and a plurality of partition members **122** cooperating with the connecting bracket **121** are disposed. Structural strength of the reinforcing bracket **120** is improved by the first reinforcing plate **1211**, so that the reinforcing bracket **120** can effectively limit and fix the battery **200**. Through cooperation between the plurality of partition members **122**, a placement space **124** is formed on the reinforcing bracket **120** to accommodate the battery cells **10**, so that one reinforcing bracket **120** can simultaneously limit at least one battery cell **10**, to resolve a technical problem that a space for disposing the battery cells **10** is reduced due to disposition of a plurality of reinforcing brackets **120**, thereby improving the energy density of the battery **200**. In addition, the positions of the battery cells **10** may be further limited through cooperation of the plurality of partition members **122**, to effectively ensure a limiting effect of the reinforcing bracket **120**, so that the reinforcing bracket **120** can limit each of the battery cells **10**, thereby increasing structural strength between the plurality of battery cells **10**.

[0060] It should be noted that, the battery **200** disclosed in the embodiments of this application may be used in but not be limited to a power consuming apparatus **1000** such as a vehicle, a ship, or an aircraft. A power supply system including the battery **200** disclosed in this application and constituting the power consuming apparatus **1000** may be used. In this way, it is conducive to improve service safety of the power supply system in the power consuming apparatus **1000**, and can improve a service life of the power supply system.

[0061] The reinforcing bracket **120** of the battery **200** in the embodiments of the present application is described below with reference to FIG. 1 to FIG. 6.

[0062] As shown in FIG. 1 and FIG. 2, the battery **200** in the embodiments of the present application includes a plurality of battery cells **10**. The plurality of battery cells **10** are used in

cooperation to improve a capacity of the battery **200**, and ensure that the battery **200** can effectively supply power to the power consuming apparatus **1000**, to ensure operating performance of the battery **200**.

[0063] It should be noted that, in the battery **200** in the present application, the battery **200** may be formed by the plurality of battery cells **10** connected in series, in parallel, or in series-parallel. Series-parallel connection means that the plurality of battery cells **10** are connected both in series and in parallel. This is not specifically limited in the present application.

[0064] As shown in FIG. 2 and FIG. 3, the reinforcing bracket **120** in the embodiments of the present application includes a connecting bracket **121** and a plurality of partition members **122**.

[0065] As shown in FIG. 3, FIG. 4, and FIG. 5, the connecting bracket **121** includes a first reinforcing plate **1211**, a plurality of avoidance holes **121a** are provided on the first reinforcing plate **1211**, the plurality of avoidance holes **121a** are spaced apart along a first direction X, each avoidance hole **121a** runs through the first reinforcing plate **1211** in a second direction Y, and the first direction X is perpendicular to the second direction Y. The first direction X described herein may be understood as the direction X shown in FIG. 3, namely, a length direction of the reinforcing bracket **120**. The second direction Y may be understood as the direction Y shown in FIG. 3, namely, a height direction of the reinforcing bracket **120**.

[0066] In other words, the plurality of avoidance holes **121a** that are spaced apart along the length direction of the reinforcing bracket **120** are provided on the first reinforcing plate **1211** in the present application and each avoidance hole **121a** runs through the first reinforcing plate **1211** in the height direction of the reinforcing bracket **120**, so that a plurality of avoidance through-holes that are in communication with two opposite sides of the first reinforcing plate **1211** are provided on the first reinforcing plate **1211**. In this way, while the battery cell **10** is fixed by the reinforcing bracket **120**, it is convenient to extend a part of a structure on the battery cell **10** through the avoidance hole **121a**, to implement an electrical connection between the battery cells **10**, that is, avoid affecting assembly of the battery **200** due to disposition of the reinforcing bracket **120**, thereby ensuring portability of assembly of the battery **200**.

[0067] In addition, the avoidance hole **121a** is provided, so that it is further conducive to reducing a material of the first reinforcing plate **1211**, reducing production costs of the first reinforcing plate **1211**, and reducing a weight of the first reinforcing plate **1211**.

[0068] It should be noted that, in this application, the reinforcing bracket **120** is disposed to include the first reinforcing plate **1211**, so that structural strength of the reinforcing bracket **120** can be improved by the first reinforcing plate **1211**, to ensure that the reinforcing bracket **120** can effectively limit the battery cells **10**, improve position stability of the battery cells **10**, and further improve structural strength between the plurality of battery cells **10**.

[0069] As shown in FIG. 3, FIG. 4, and FIG. 6, the plurality of partition members **122** are spaced apart on the connecting bracket **121** along the first direction X. A placement space **124** is defined between adjacent partition members **122** and the connecting bracket **121**. The placement space **124** is used for placing the battery cells **10**. The plurality of partition members **122** are spaced apart on the connecting bracket **121** along the length direction of the reinforcing bracket **120**. In addition, after the plurality of partition members **122** and the connecting bracket **121** are assembled, the placement space **124** used for placing the battery cells **10** is defined between the adjacent partition members **122** and the connecting bracket **121**, and the battery cells **10** are placed in the placement space **124**, to fix the battery cells **10** by the reinforcing brackets **120**. In addition, the battery cells **10** may be further limited by the adjacent partition members **122**, to prevent the battery cells **10** from being displaced in the first direction X, thereby improving the position stability of the battery cells **10**.

[0070] In addition, the plurality of partition members **122** are disposed on the reinforcing bracket **120** in the present application, and one placement space **124** can be defined between each two adjacent partition members **122**. In this way, in a process of using the reinforcing bracket **120**, a

quantity of placement spaces **124** can be properly designed according to a quantity of to-be-fixed battery cells **10**, to ensure that one reinforcing bracket **120** can simultaneously support a plurality of battery cells **10**, so as to reduce a quantity of used reinforcing brackets **120**. In this way, while a cost of using the reinforcing bracket **120** is reduced, a space occupied by the reinforcing bracket **120** can be further reduced, to resolve the technical problem in the related art that a space for disposing the battery cells **10** is reduced due to disposition of the plurality of reinforcing brackets **120**, thereby improving the energy density of the battery **200**.

[0071] Therefore, it can be learnt from the foregoing structure that, the connecting bracket **121** including the first reinforcing plate **1211** is disposed in the reinforcing bracket **120** of the battery **200** in the present application, to improve the structural strength of the reinforcing bracket **120** by the first reinforcing plate **1211**, thereby ensuring that the battery cells **10** can be effectively limited by the reinforcing brackets **120**, improving position stability of the battery cells **10**, and increasing structural strength between the plurality of battery cells **10**.

[0072] In addition, the reinforcing bracket **120** is disposed to include the plurality of partition members **122** that are spaced apart along the first direction X. In this way, after the plurality of partition members **122** and the connecting bracket **121** are assembled, the reinforcing bracket **120** including the placement space **124** may be formed, and the battery cells **10** are placed in the placement space **124**. The plurality of battery cells **10** may be simultaneously fixed by the reinforcing brackets **120**, to resolve the technical problem that the space for disposing the battery cells **10** is reduced due to disposition of the plurality of reinforcing brackets **120**, thereby improving the energy density of the battery **200**. In addition, corresponding battery cells **10** may be further limited and fixed by the plurality of partition members **122**, to prevent the battery cells **10** from being displaced in the first direction X, thereby further improving the position stability of the battery cells **10**.

[0073] It should be further noted that, the placement space **124** is formed in the reinforcing bracket **120**, so that during assembly of the reinforcing bracket **120** and the battery cells **10**, the battery cells **10** may be directly disposed in the placement space **124** in a manner of cooperation. With such a structural design, a structural member connecting the reinforcing bracket **120** and the battery cells **10** may be omitted, which not only helps reduce a weight of the entire battery **200**, but also simplifies an assembly process, to reduce difficulty in assembling and disassembling the reinforcing bracket **120** and the battery cells **10**, and further reduce production costs.

[0074] It can be learnt from the above that, while the battery cells **10** are fixed by the reinforcing bracket **120** in this application, the position stability of the battery cells **10** can be effectively improved and structural strength between the plurality of battery cells **10** can be increased, difficulty in assembly between the reinforcing bracket **120** and the battery cells **10** can be further reduced, and the energy density of the battery **200** can be ensured.

[0075] It may be understood that, compared with the related art, in the present application, the first reinforcing plate **1211** is disposed on the reinforcing bracket **120** and the reinforcing bracket **120** includes the plurality of placement spaces **124**. While the position stability of the battery cells **10** is improved and the structural strength among the plurality of battery cells **10** is increased by the reinforcing bracket **120**, it is further conducive to ensure the energy density of the battery **200**.

[0076] Optionally, the plurality of partition members **122** are fixedly connected to the connecting bracket **121**. While the plurality of partition members **122** are disposed on the connecting bracket **121**, connection strength between the plurality of partition members **122** and the connecting bracket **121** may be further ensured. In this way, this helps support the plurality of partition members **122** by the connecting bracket **121**, to improve position stability of a plurality of partition members **122**, so that the plurality of placement spaces **124** having stable structures are formed to accommodate the battery cells **10**.

[0077] It should be noted that, the fixed connection described herein may be non-detachable connection such as soldering connection or adhesive connection, or may be detachable connection

such as bolt connection or clamping connection. This is not specifically limited in the present application, provided that it is ensured that the plurality of partition members **122** can be stably connected to the connecting bracket **121**.

[0078] Certainly, in some other embodiments, the plurality of partition members **122** may also be integrally formed on the connecting bracket **121**, to ensure the connection strength between the plurality of partition members **122** and the connecting bracket **121**.

[0079] Optionally, the first reinforcing plate **1211** is a steel plate, an aluminum plate, or a glass fiber plate, to ensure structural strength of the first reinforcing plate **1211**, so as to improve the structural strength of the reinforcing bracket **120** by the first reinforcing plate **1211**.

[0080] In addition, through the foregoing disposition, displacement of the battery cells **10** may be further limited by the first reinforcing plate **1211**, to further improve the position stability of the battery cells **10**.

[0081] In a specific example, the first reinforcing plate **1211** is mainly configured to limit positions of the battery cells **10** in the second direction Y.

[0082] In some embodiments, as shown in FIG. 1 and FIG. 2, the battery **200** includes a plurality of groups of battery units **10A**, each group of battery units **10A** includes a plurality of battery cells **10**, and each reinforcing bracket **120** is disposed corresponding to one group of battery units **10A**. In this way, the plurality of battery cells **10** in a group of battery units **10A** are simultaneously limited by one reinforcing bracket **120**, and a quantity of used reinforcing brackets **120** is further reduced and the energy density of the battery **200** is improved while structural strength of the battery units **10A** is increased.

[0083] In some embodiments, as shown in FIG. 2, FIG. 3, and FIG. 4, the reinforcing bracket **120** further includes a second reinforcing plate **123**. The second reinforcing plates **123** are disposed on at least one side of each placement space **124** in a third direction Z, and the third direction Z is separately disposed perpendicular to the first direction X and the second direction Y. The third direction Z described herein may be understood as the direction Z shown in FIG. 3, namely, a width direction of the reinforcing bracket **120**.

[0084] Therefore, the above may also be understood as that, second reinforcing plates **123** are disposed on at least one side of each placement space **124** in the width direction of the reinforcing bracket **120**. In this case, the battery cells **10** in the placement space **124** may be limited and fixed by the second reinforcing plates **123**, to improve the position stability of the battery cells **10** in the placement space **124**. In addition, the structural strength of the reinforcing bracket **120** may be further improved by the second reinforcing plates **123**, to improve limiting quality of the reinforcing bracket **120**.

[0085] In other words, the second reinforcing plates **123** in the present application may be configured to limit the battery cells **10** in the placement space **124**. In addition, the structural strength of the reinforcing bracket **120** may be further improved, so that the reinforcing bracket **120** can stably limit the battery cells **10**, thereby improving the position stability of the battery cells **10**, increasing the structural strength between the plurality of battery cells **10**, and improving reliability of the battery **200**.

[0086] In a specific example, the first reinforcing plate **1211** is mainly configured to limit positions of the battery cells **10** in the third direction Z.

[0087] Optionally, as shown in FIG. 3 and FIG. 4, the second reinforcing plate **123** is disposed on the partition member **122**, to support the second reinforcing plates **123** by the partition member **122**, and position stability of the second reinforcing plates **123** is improved, so that the structural strength of the reinforcing bracket **120** is improved by the second reinforcing plate **123**, and the positions of the battery cells **10** are limited by the second reinforcing plates **123**.

[0088] Optionally, the second reinforcing plates **123** are disposed on the partition member **122** in a connection manner such as soldering connection, adhesive connection, or bolting connection, to implement a fixed connection between the second reinforcing plates **123** and the partition member

122, ensure connection strength between the second reinforcing plates **123** and the partition member **122**, and improve the position stability of the second reinforcing plates **123**.

[0089] Optionally, the second reinforcing plates **123** are steel plates, aluminum plates, or glass fiber plates, to ensure that the second reinforcing plates **123** have specific structural strength, so as to improve the structural strength of the reinforcing bracket **120** by the second reinforcing plates **123** and ensure that displacement of the battery cells **10** can be effectively limited by the second reinforcing plates **123**.

[0090] In some embodiments, as shown in FIG. 3 and FIG. 4, in the third direction Z, the second reinforcing plates **123** are disposed on two sides of each placement space **124**. In this way, while the battery cells **10** are disposed in the placement space **124**, the battery cells **10** may be limited through cooperation between the second reinforcing plates **123** on two opposite sides of the placement space **124**, to effectively avoid displacement of the battery cells **10** in the third direction Z, thereby improving the position stability of the battery cells **10** in the placement space **124**, and improving the structural strength among the plurality of battery cells **10**.

[0091] In addition, through the foregoing disposition, the reinforcing bracket **120** in the present application may further include a plurality of second reinforcing plates **123**. The plurality of second reinforcing plates **123** cooperate with each other to further improve the structural strength of the reinforcing bracket **120**, and ensure a reinforcing effect and a limiting effect of the reinforcing bracket **120**.

[0092] In some embodiments, as shown in FIG. 3 and FIG. 4, in the first direction X, the second reinforcing plates **123** separately protrude from two sides of the same partition member **122**. In this way, one second reinforcing plate **123** may be simultaneously disposed on a same side portion of two adjacent placement spaces **124**, to achieve an objective of simultaneously limiting and fixing two battery cells **10** by one second reinforcing plate **123**. In this way, while the position stability of the battery cells **10** is improved, a quantity of disposed second reinforcing plates **123** may be further reduced, to reduce production costs and manufacturing difficulty of the reinforcing bracket **120**.

[0093] Optionally, as shown in FIG. 3 and FIG. 4, the second reinforcing plates **123** are disposed on the partition member and two opposite ends of the second reinforcing plate **123** in the first direction X respectively extend in a direction away from each other, so that in the first direction X, the second reinforcing plates **123** can separately protrude from two sides of the same partition member **122**, to achieve an objective of simultaneously limiting two battery cells **10** by one second reinforcing plate **123**.

[0094] Optionally, as shown in FIG. 2, a plurality of side walls **11** are disposed on each battery cell **10**, the plurality of side walls **11** include a first side wall **111** and a second side wall **112**, the first side wall **111** is connected to the second side wall **112**, and the first side wall **111** is a side wall **11** having a largest area of the battery cell **10**. It may also be understood herein that, each battery cell **10** includes the first side wall **111** and the second side wall **112** that are connected to each other. An area of the second side wall **112** is less than an area of the first side wall **111**, so that the first side wall **111** is a “large surface” of the battery cell **10**.

[0095] Optionally, as shown in FIG. 2 and FIG. 3, the second reinforcing plates **123** are respectively disposed on two sides of the placement space **124** and abut against and cooperate with the first side wall **111** of the battery cell **10**, so that displacement of the corresponding battery cell **10** in the third direction Z is limited by the second reinforcing plates **123**.

[0096] It should be noted that, because the first side wall **111** is the “large surface” of the battery cell **10**, the second reinforcing plates **123** are disposed to cooperate with the first side wall **111** of the battery cell **10**, so that by increasing a cooperation area between the second reinforcing plates **123** and the first side wall **111**, limiting quality and reinforcing quality of the reinforcing bracket **120** can be improved, to effectively improve position stability of the battery cell **10** and increase the structural strength between the plurality of battery cells **10** by the reinforcing bracket **120**.

[0097] In some embodiments, as shown in FIG. 3 and FIG. 4, in the second direction Y, a height of the second reinforcing plate **123** is less than a height of the partition member **122**. In other words, in the height direction of the reinforcing bracket **120**, the height of the second reinforcing plate **123** is less than the height of the partition member **122**. In this way, while ensuring a reinforcing function of the second reinforcing plates **123**, materials of the second reinforcing plates **123** can be further reduced, so as to reduce manufacturing costs of the reinforcing bracket **120** and further reduce difficulty in forming the reinforcing bracket **120**.

[0098] In addition, the height of the second reinforcing plate **123** is set to be less than the height of the partition member **122**, so that assembly difficulty between the second reinforcing plates **123** and the partition member **122** may be further reduced, to ensure that the second reinforcing plates **123** can be effectively disposed on the partition member **122**, so as to achieve an objective that the partition member **122** supports the second reinforcing plates **123**, and improve the position stability of the second reinforcing plates **123**.

[0099] Certainly, in some other examples, in the second direction Y, the height of the second reinforcing plate **123** may also be set to be equal to the height of the partition member **122**. In this way, while ensuring that the second reinforcing plates **123** can be effectively disposed on the partition member **122**, it can be further ensured that the second reinforcing plates **123** have a sufficient area, so that the structural strength of the reinforcing bracket **120** is improved by the second reinforcing plates **123**, and a cooperation area between the second reinforcing plates **123** and the first side wall **111** is increased. In this way, displacement of the battery cell **10** can be effectively limited by the second reinforcing plates **123**.

[0100] In some embodiments, as shown in FIG. 3 and FIG. 4, second reinforcing plates **123** and the connecting bracket **121** are spaced apart, to further reduce difficulty in forming the reinforcing bracket **120** and improve processing efficiency of the reinforcing bracket **120**.

[0101] In addition, because the height of the second reinforcing plate **123** is less than the height of the partition member **122**, after the second reinforcing plates **123** and the connecting bracket **121** are spaced apart, the second reinforcing plates **123** can be disposed close to the middle of the partition member **122** in a height direction. In this way, while the second reinforcing plates **123** abut against and cooperate with the first side wall **111** of the battery cell **10**, the second reinforcing plates **123** can abut against at a position in the middle of the first side wall **111** in the second direction Y, to ensure that a structure in which the first side wall **111** is located on an upper part of the second direction Y and a structure in which the first side wall **111** is located on a lower part of the second direction Y are uniformly limited, and ensure that displacement of the battery cells **10** in the third direction Z can be effectively limited by the second reinforcing plates **123**.

[0102] In a specific example, as shown in FIG. 3 and FIG. 4, in a height direction of the reinforcing bracket **120**, the height of the second reinforcing plate **123** is less than the height of the partition member **122**, and in a process of connecting the second reinforcing plates **123** and the partition member **122**, the second reinforcing plates **123** and the connecting bracket **121** are spaced apart, to reduce assembly difficulty of the reinforcing bracket **120**, improve assembly efficiency, and ensure a limiting effect of the reinforcing bracket **120**.

[0103] Certainly, in some other examples, the second reinforcing plates **123** may also be connected to the connecting bracket **121**, to support the second reinforcing plates **123** by the connecting bracket **121**, thereby improving position stability of the second reinforcing plates **123**.

[0104] In some embodiments, as shown in FIG. 3 and FIG. 4, a through-hole **1221** is provided on each partition member **122**, and the through-hole **1221** runs through the partition member **122** in the first direction X. In this way, the through-hole running through the partition member **122** is formed on the partition member **122**. The through-hole is provided, so that manufacturing costs of the partition member **122** can be effectively reduced, that is, manufacturing costs of the reinforcing bracket **120** can be reduced, and lightweight of the reinforcing bracket **120** is also implemented.

[0105] In addition, in a process of assembling the battery cells **10** and the reinforcing bracket **120**,

two adjacent battery cells **10** are located on two opposite sides of the partition member **122** in the first direction X. The through-hole **1221** is provided on the partition member **122**, so that it is convenient to connect the two adjacent battery cells **10** through the through-hole **1221**, to improve connection strength between the adjacent battery cells **10** and ensure reliability of the battery **200**. In addition, it is further conducive to accelerate heat dissipation of the battery cells **10** and ensure normal expansion between the adjacent battery cells **10**, thereby improving service safety of the battery cells **10**.

[0106] Optionally, in the first direction X, the two adjacent battery cells **10** are connected by a heat conduction adhesive. While the two adjacent battery cells **10** are fixedly connected, a heat dissipation effect of the battery cells **10** may be further improved.

[0107] In some embodiments, as shown in FIG. 3 and FIG. 4, the connecting bracket **121** includes an annular enclosing plate **1212**. The enclosing plate **1212** is connected to an outer edge of the first reinforcing plate **1211**. The enclosing plate **1212** extends along the second direction Y. In other words, the annular enclosing plate **1212** extends along the height direction of the reinforcing bracket **120**, so that the enclosing plate **1212** has a specific extension area, thereby improving structural strength of the reinforcing bracket **120** by the enclosing plate **1212**.

[0108] In addition, because the enclosing plate **1212** is connected to the outer edge of the first reinforcing plate **1211**, after the enclosing plate **1212** is disposed to extend along the second direction Y, the enclosing plate **1212** may be extended to an outer side of the placement space **124**, to limit displacement of the battery cells **10** by the enclosing plate **1212**, thereby improving the position stability of the battery cells **10**.

[0109] In a specific example, a part of the enclosing plate **1212** is located on two opposite sides of the placement space **124** in the third direction Z, so that displacement of the battery cell **10** in the third direction Z is limited by the enclosing plate **1212**, and another part of the enclosing plate **1212** is located on an outer side of two outermost placement spaces **124** in the first direction X, to limit displacement of the entire battery unit **10A** in the first direction X by the enclosing plate **1212**.

[0110] In other words, displacement of a single battery cell **10** in the first direction X may be limited through cooperation between the two adjacent partition members **122**, and the displacement of the entire battery unit **10A** in the first direction X may be limited by the enclosing plate **1212**, to improve structural stability and structural strength of the battery unit **10A**.

[0111] Optionally, the enclosing plate **1212** is a steel plate, an aluminum plate, or a glass fiber plate, to ensure that the enclosing plate **1212** has specific structural strength, thereby ensuring that the displacement of the battery cell **10** can be effectively limited by the enclosing plate **1212**.

[0112] Optionally, as shown in FIG. 3, in the second direction Y, the enclosing plate **1212** and the second reinforcing plates **123** are spaced apart, to further reduce difficulty in forming the reinforcing bracket **120** and improve processing efficiency of the reinforcing bracket **120**.

[0113] Certainly, in some other examples, the enclosing plate **1212** may also be connected to the second reinforcing plates **123**, so that relative position stability of the enclosing plate **1212** and the second reinforcing plates **123** is ensured, thereby improving the limiting quality of the reinforcing bracket **120**.

[0114] In some embodiments, as shown in FIG. 3 and FIG. 5, the reinforcing bracket **120** is symmetrically disposed along a first center line extending parallel to the first direction X. An extension direction of the first center line is parallel to the first direction X, and the reinforcing bracket **120** is symmetrically disposed along the first center line, so that the reinforcing bracket **120** is formed as a symmetric member symmetrically disposed along the first center line, that is, structures of the reinforcing brackets **120** located on the two opposite sides of the third direction Z are symmetrically disposed, to reduce processing difficulty of the reinforcing bracket **120**, and further ensure uniform reinforcing strength and limiting strength of the reinforcing bracket **120** on the plurality of battery cells **10** in the third direction Z, thereby improving the position stability of the plurality of battery cells **10**.

[0115] In some embodiments, as shown in FIG. 3 and FIG. 6, the reinforcing bracket **120** is symmetrically disposed along a second center line parallel to the second direction Y. An extension direction of the second center line is parallel to the second direction Y, and the reinforcing bracket **120** is symmetrically disposed along the second center line, so that the reinforcing bracket **120** is formed as a symmetric member symmetrically disposed along the second center line, that is, structures of the reinforcing brackets **120** located on the two opposite sides of the first direction X are symmetrically disposed, to further reduce processing difficulty of the reinforcing bracket **120**, and further ensure uniform reinforcing strength and limiting strength of the reinforcing bracket **120** on the plurality of battery cells **10** in the first direction X, thereby improving the position stability of the plurality of battery cells **10**.

[0116] In conclusion, it can be learned from the above that, the reinforcing bracket **120** in this application is not only symmetrically disposed along the first center line extending parallel to the first direction X, but also symmetrically disposed along the second center line extending parallel to the second direction Y, to maximally reduce the processing difficulty of the reinforcing bracket **120**, and improve the reinforcing quality of the reinforcing bracket **120**.

[0117] The following describes the reinforcing bracket **120** of the battery **200** in this application in detail with reference to the accompanying drawings of the specification.

[0118] As shown in FIG. 3 and FIG. 4, the reinforcing bracket **120** is symmetrically disposed along the first center line extending parallel to the first direction X and is symmetrically disposed along the second center line extending parallel to the second direction Y, and the reinforcing bracket **120** includes: a connecting bracket **121**, a partition member **122**, and second reinforcing plates **123**.

[0119] With reference to FIG. 3, FIG. 4, and FIG. 5, the connecting bracket **121** includes a first reinforcing plate **1211** and an annular enclosing plate **1212**. A plurality of avoidance holes **121a** are provided on the first reinforcing plate **1211**. The plurality of avoidance holes **121a** are spaced apart along the first direction X and each of the avoidance holes **121a** runs through the first reinforcing plate **1211** in the second direction Y. The enclosing plate **1212** is connected to the outer edge of the first reinforcing plate **1211** and the enclosing plate **1212** extends along the second direction Y.

[0120] As shown in FIG. 3, FIG. 4, and FIG. 6, a plurality of partition members **122** are spaced apart on the connecting bracket **121** along the first direction X. A through-hole **1221** is provided on each partition member **122**. The through-hole **1221** runs through the partition member **122** in the first direction X. A placement space **124** is defined between adjacent partition members **122** and the connecting bracket **121**. The placement space **124** is used for placing the battery cells **10**.

[0121] As shown in FIG. 3 and FIG. 4, second reinforcing plates **123** are disposed on two opposite sides of each placement space **124** in the third direction Z, and in the first direction X, the second reinforcing plates **123** separately protrude from two sides of the same partition member **122**, so that one second reinforcing plate **123** is simultaneously located on side portions of two adjacent placement spaces **124**.

[0122] In addition, as shown in FIG. 3 and FIG. 4, in the second direction Y, the height of the second reinforcing plate **123** is less than the height of the partition member **122**, so that the second reinforcing plates **123** and the connecting bracket **121** are spaced apart.

[0123] A battery **200** in the embodiment of the second aspect in the present application is described below with reference to the accompanying drawings in this specification.

[0124] As shown in FIG. 1 and FIG. 2, the battery **200** in the embodiments of the present application includes a reinforcing bracket **120** and a plurality of battery cells **10**.

[0125] As shown in FIG. 2, the reinforcing bracket **120** is the foregoing reinforcing bracket **120**, and a specific structure of the reinforcing bracket **120** is not described herein again.

[0126] As shown in FIG. 2 and FIG. 3, battery cells **10** are placed in each placement space **124**, the first reinforcing plate **1211** is located on an outer side of the corresponding battery cells **10**, and an electrode terminal **12** of each of the battery cells **10** passes through a corresponding avoidance hole **121a**. While the reinforcing bracket **120** in the present application is used in the battery **200**, the

battery cells **10** are placed in each placement space **124**, and the first reinforcing plate **1211** is located on the outer side of the corresponding battery cells **10**, so that partition members **122** on two sides of the placement space **124** and the first reinforcing plate **1211** cooperate with each other to limit the battery cells **10**, to improve position stability of the battery cells **10** and ensure structural strength between adjacent battery cells **10**.

[0127] In addition, while the battery cells **10** are placed in the placement space **124**, the electrode terminal **12** of each of the battery cells **10** further passes through the corresponding avoidance hole **121a**, to prevent the reinforcing bracket **120** from hindering cooperative connection between the electrode terminals **12**.

[0128] In other words, in the present application, while structural strength of the reinforcing bracket **120** is improved by the first reinforcing plate **1211**, the first reinforcing plate **1211** may further avoid the electrode terminals **12** of the battery cells **10**, so that the electrode terminals **12** of the battery cells **10** can extend out of the corresponding avoidance hole **121a**, to facilitate an electrical connection between the battery cells **10**, reduce assembly difficulty of the battery **200**, and improve assembly efficiency.

[0129] It should be noted that, because the reinforcing bracket **120** in the embodiments of the present application has the foregoing technical effect, the battery **200** in the embodiments of the present application also has the foregoing technical effect. That is, the reinforcing bracket **120** in the present application is used, so that structural stability of the battery **200** and reliability of the battery **200** can be effectively ensured, and an energy density of the battery **200** can be improved.

[0130] Optionally, the electrode terminal **12** and a peripheral wall of the avoidance hole **121a** are spaced apart, to avoid interference between the reinforcing bracket **120** and the electrode terminal **12**, thereby facilitating improving an insulation property between the reinforcing bracket **120** and the electrode terminal **12**.

[0131] Optionally, the avoidance hole **121a** is further used for avoiding a pressure relief structure of the battery cell **10**. The pressure relief structure is disposed on the battery cell **10**. While the battery cell **10** is placed in the placement space **124**, the pressure relief structure of each battery cell **10** may also pass through a corresponding avoidance hole **121a**, to prevent the reinforcing bracket **120** from hindering pressure relief of the pressure relief structure, that is, ensure that the battery cell **10** can effectively perform pressure relief by the pressure relief structure, thereby improving service safety of the battery cell **10**, that is, ensuring service safety of the battery **200**.

[0132] In some examples, as shown in FIG. 1 and FIG. 2, the battery **200** includes a plurality of reinforcing brackets **120** and a plurality of groups of battery units **10A**. Each group of battery units **10A** includes a plurality of battery cells **10**, and the plurality of reinforcing brackets **120** are in one-to-one correspondence with the plurality of groups of battery units **10A**, so that the plurality of battery cells **10** in the group of battery units **10A** are simultaneously supported by one reinforcing bracket **120**, and an energy density of the battery **200** can be further improved while the structural strength of the battery **200** is improved.

[0133] In some embodiments, an adhesive layer is disposed between the reinforcing bracket **120** and each of the battery cells **10**. The adhesive layer is used to implement a fixed connection between the reinforcing bracket **120** and each of the battery cells **10**, to ensure that the battery cells **10** can be stably disposed in the placement space **124** of the reinforcing bracket **120**, improve the position stability of the battery cells **10** relative to the reinforcing bracket **120**, and achieve an objective of limiting and reinforcing the battery cells **10** by the reinforcing bracket **120**, thereby ensuring structural strength between the adjacent battery cells **10**.

[0134] Optionally, the adhesive layer is formed by a heat conduction adhesive. In this way, while the reinforcing bracket **120** is fixedly connected to the battery cells **10** by the adhesive layer, the adhesive layer may be further used to conduct heat generated during operation of the battery cells **10**, to improve efficiency of heat dissipation of the battery cells **10**, thereby ensuring service safety of the battery cells **10**, and prolonging the service life of the battery cells **10**.

[0135] In some embodiments, as shown in FIG. 1 and FIG. 2, the battery **200** further includes a box **30**, a plurality of battery cells **10** are disposed in the box **30**, and the reinforcing bracket **120** is fixed to the box **30**. In other words, both the plurality of battery cells **10** and the reinforcing brackets **120** are disposed in the box **30**, so that the reinforcing brackets **120** are disposed close to positions of the plurality of battery cells **10**, thereby facilitating limiting the plurality of battery cells **10** by the reinforcing bracket **120**. In addition, the box **30** may be further used to support the reinforcing bracket **120** and the plurality of battery cells **10**, so that relative position stability between the reinforcing bracket **120** and the plurality of battery cells **10** is improved, that is, the limiting effect of the reinforcing bracket **120** is improved.

[0136] In some examples, an accommodating cavity is formed in the box **30**, and the plurality of battery cells **10** and the reinforcing brackets **120** are all disposed in the accommodating cavity, so that the plurality of battery cells **10** and the reinforcing brackets **120** are disposed in the box **30**.

[0137] The box **30** may use a plurality of structures.

[0138] In some embodiments, the box **30** may include a first portion and a second portion. The second portion may be a hollow structure with one end open, the first portion may be a plate-shaped structure, and the first portion covers an opening side of the second portion, so that the first portion and the second portion jointly define the accommodating cavity accommodating the battery cells **10** and the reinforcing brackets **120**. Alternatively, both the first portion and the second portion may also be hollow structures with one side open, and an opening side of the first portion covers an opening side of the second portion, so that the first portion and the second portion jointly define the accommodating cavity accommodating the battery cells **10** and the reinforcing brackets **120**. Certainly, the box **30** formed by the first portion and the second portion may have various shapes, such as a cylinder, a cuboid, and the like.

[0139] Optionally, as shown in FIG. 1, FIG. 2, and FIG. 3, the battery cells **10** are disposed in the placement space **124** of the reinforcing bracket **120** and are located between a bottom wall of the box **30** and the first reinforcing plate **1211** in the second direction Y. In this case, displacement of the battery cells **10** in the second direction Y may be limited through cooperation between the bottom wall of the box **30** and the first reinforcing plate **1211**, to limit the battery cells **10** by the reinforcing bracket **120**.

[0140] Optionally, as shown in FIG. 1, FIG. 2, and FIG. 3, the battery cells **10** are respectively disposed on two sides of each partition member **122** in the first direction X, and an outermost battery cell **10** of the battery unit **10A** in the first direction X is disposed between one of the partition members **122** and an inner wall of the box **30**. In this way, displacement of the outermost battery cell **10** of the battery unit **10A** in the first direction X is limited through cooperation between the partition member **122** and the inner wall of the box **30**, thereby improving structural strength of the battery unit **10A**.

[0141] In conclusion, the reinforcing bracket **120** in this application may be configured to limit displacement of each of the battery cells **10** in the first direction X, the second direction Y, and the third direction Z, prevent slippage of the battery cells **10** in the first direction X, the second direction Y, and the third direction Z, improve the position stability of the battery cells **10**, ensure the structural strength between the adjacent battery cells **10**, and increase the structural stability of the battery **200**.

[0142] The following describes the power consuming apparatus **1000** in the embodiment in the third aspect in the present application with reference to the accompanying drawings in the specification.

[0143] As shown in FIG. 7, the power consuming apparatus **1000** in this embodiment of the present application includes the battery **200** in the foregoing embodiment, and the battery **200** is configured to provide electric energy.

[0144] Because the battery **200** in this embodiment of the present application has the foregoing technical effect, the power consuming apparatus **1000** in this embodiment of the present application

also has the foregoing technical effect. That is, structural stability of the power consuming apparatus **1000** can be effectively improved and structural strength of the power consuming apparatus **1000** can be ensured by the battery **200** in the present application.

[0145] It should be noted that, the embodiment in the third aspect in the present application provides a power consuming apparatus **1000** using the battery **200** as a power supply. The power consuming apparatus **1000** may include but is not limited to a mobile phone, a tablet, a notebook computer, an electric toy, an electric tool, an electric scooter, an electric vehicle, a vehicle, a ship, a spacecraft, or the like. The electric toy may include a fixed or mobile electric toy, such as a game console, an electric vehicle toy, an electric ship toy, or an electric airplane toy. The spacecraft may include an airplane, a rocket, a space shuttle, a spaceship, or the like.

[0146] As shown in FIG. 7, while the power consuming apparatus **1000** is the vehicle, the vehicle may be a fuel vehicle, a gas vehicle, or a new energy vehicle. The new energy vehicle may be a pure electric vehicle, a hybrid electric vehicle, an extended-range electric vehicle, or the like. The battery **200** is disposed inside the vehicle, and the battery **200** may be disposed at a bottom, head, or tail of the vehicle. The battery **200** may be configured to provide electric energy for the vehicle. For example, the battery **200** may be used as an operating power supply for the vehicle. The vehicle may further include a controller and a motor, where the controller is used for controlling the battery **200** to provide electric energy for the motor to be used for, for example, operating electricity requirements during start-up, navigation and running of the vehicle.

[0147] In some embodiments of the present application, the battery **200** may not only serve as the operating power supply for the vehicle, but also serve as a driving power supply for the vehicle, to replace or partially replace fuel or natural gas to provide driving power for the vehicle.

[0148] It may be understood that, other configurations and operations of the reinforcing bracket **120** for a battery **200**, the battery **200**, and the power consuming apparatus **1000** according to the embodiments of the present application are known to a person of ordinary skill in the art, and will not be described in detail herein.

[0149] It should be noted that the embodiments in the present application and features in the embodiments may be combined with each other without conflicts.

[0150] The foregoing descriptions are merely preferred embodiments of the present application, and are not intended to limit the present application. For a person skilled in the art, the present application may have a variety of modifications and changes. Any modification, equivalent replacement, or improvement made without departing from the spirit and principle of the present application shall fall within the protection scope of the present application.

Claims

1. A reinforcing bracket for a battery, wherein the battery comprises a plurality of battery cells, and the reinforcing bracket comprises: a connecting bracket, wherein the connecting bracket comprises a first reinforcing plate, a plurality of avoidance holes that are spaced apart along a first direction are provided on the first reinforcing plate, each of the avoidance holes runs through the first reinforcing plate in a second direction, and the first direction is perpendicular to the second direction; and a plurality of partition members, wherein the plurality of partition members are spaced apart on the connecting bracket along the first direction, and a placement space for placing the battery cells is defined between the adjacent partition members and the connecting bracket.
2. The reinforcing bracket for a battery according to claim 1, further comprising: second reinforcing plates, wherein the second reinforcing plates are disposed on at least one side of each placement space in a third direction, and the third direction is separately disposed perpendicular to the first direction and the second direction.
3. The reinforcing bracket for a battery according to claim 2, wherein the second reinforcing plates are disposed on two sides of each placement space in the third direction.

- 4.** The reinforcing bracket for a battery according to claim 2, wherein the second reinforcing plates separately protrude from two sides of the same partition member in the first direction.
 - 5.** The reinforcing bracket for a battery according to claim 2, wherein a height of the second reinforcing plate is less than a height of the partition member in the second direction.
 - 6.** The reinforcing bracket for a battery according to claim 5, wherein the second reinforcing plates and the connecting bracket are spaced apart.
 - 7.** The reinforcing bracket for a battery according to claim 1, wherein a through-hole is provided on each partition member, and the through-hole runs through the partition member in the first direction.
 - 8.** The reinforcing bracket for a battery according to claim 1, wherein the connecting bracket comprises an annular enclosing plate, the enclosing plate is connected to an outer edge of the first reinforcing plate, and the enclosing plate extends along the second direction.
 - 9.** The reinforcing bracket for a battery according to claim 1, wherein the reinforcing bracket is symmetrically disposed along a first center line extending parallel to the first direction.
 - 10.** The reinforcing bracket for a battery according to claim 9, wherein the reinforcing bracket is symmetrically disposed along a second center line parallel to the second direction.
 - 11.** A battery, comprising: a reinforcing bracket, wherein the reinforcing bracket is the reinforcing bracket according to claim 1; and a plurality of battery cells, wherein the battery cells are placed in each placement space, a first reinforcing plate is located on an outer side of the corresponding battery cell, and an electrode terminal of each of the battery cells passes through a corresponding avoidance hole.
 - 12.** The battery according to claim 11, wherein an adhesive layer is disposed between the reinforcing bracket and each of the battery cells.
 - 13.** The battery according to claim 11, further comprising a box, wherein the plurality of battery cells are disposed in the box, and the reinforcing bracket is fixed to the box.
 - 14.** A power consuming apparatus, comprising the battery according to claim 11, wherein the battery is configured to supply electric energy.
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